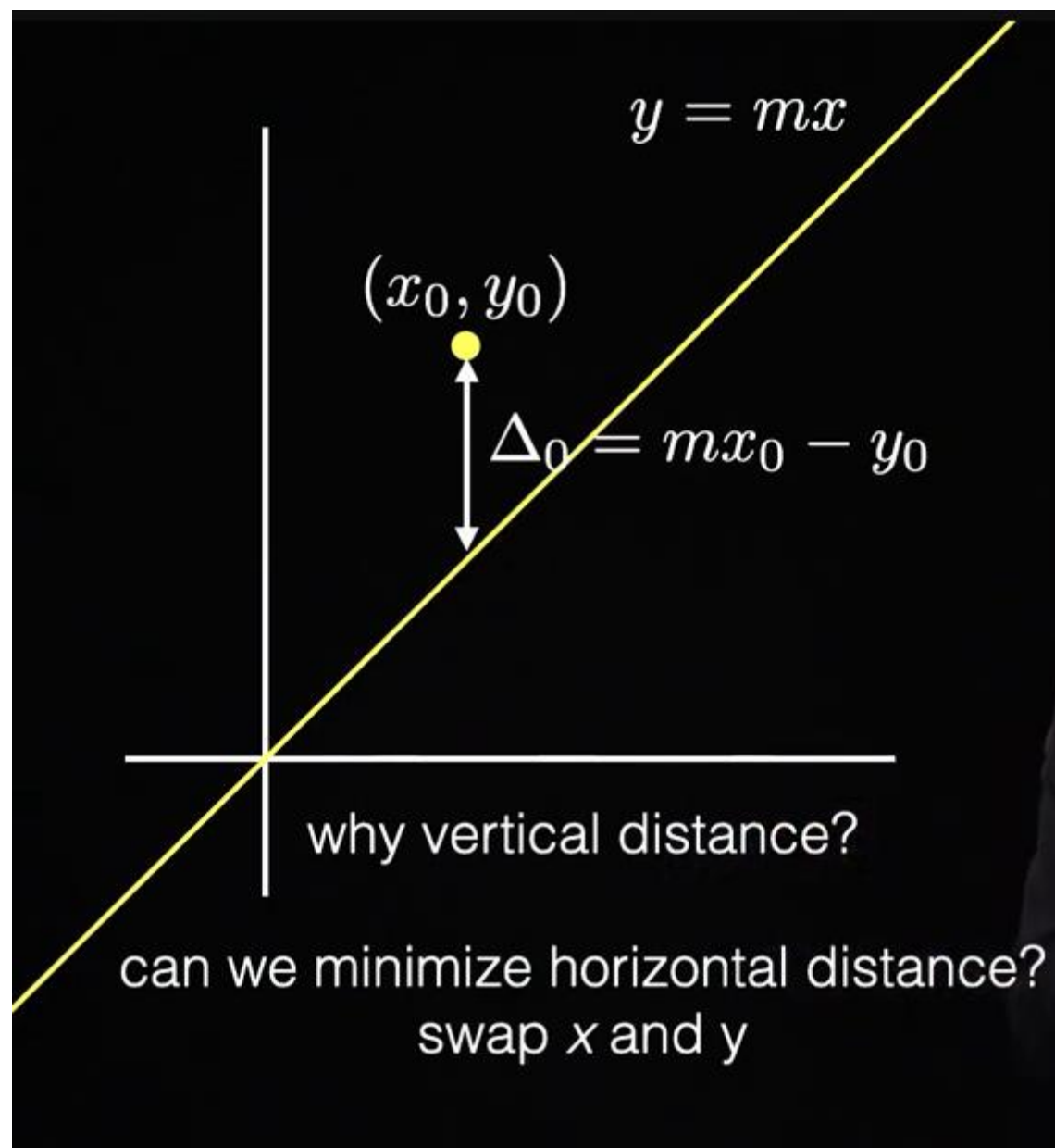
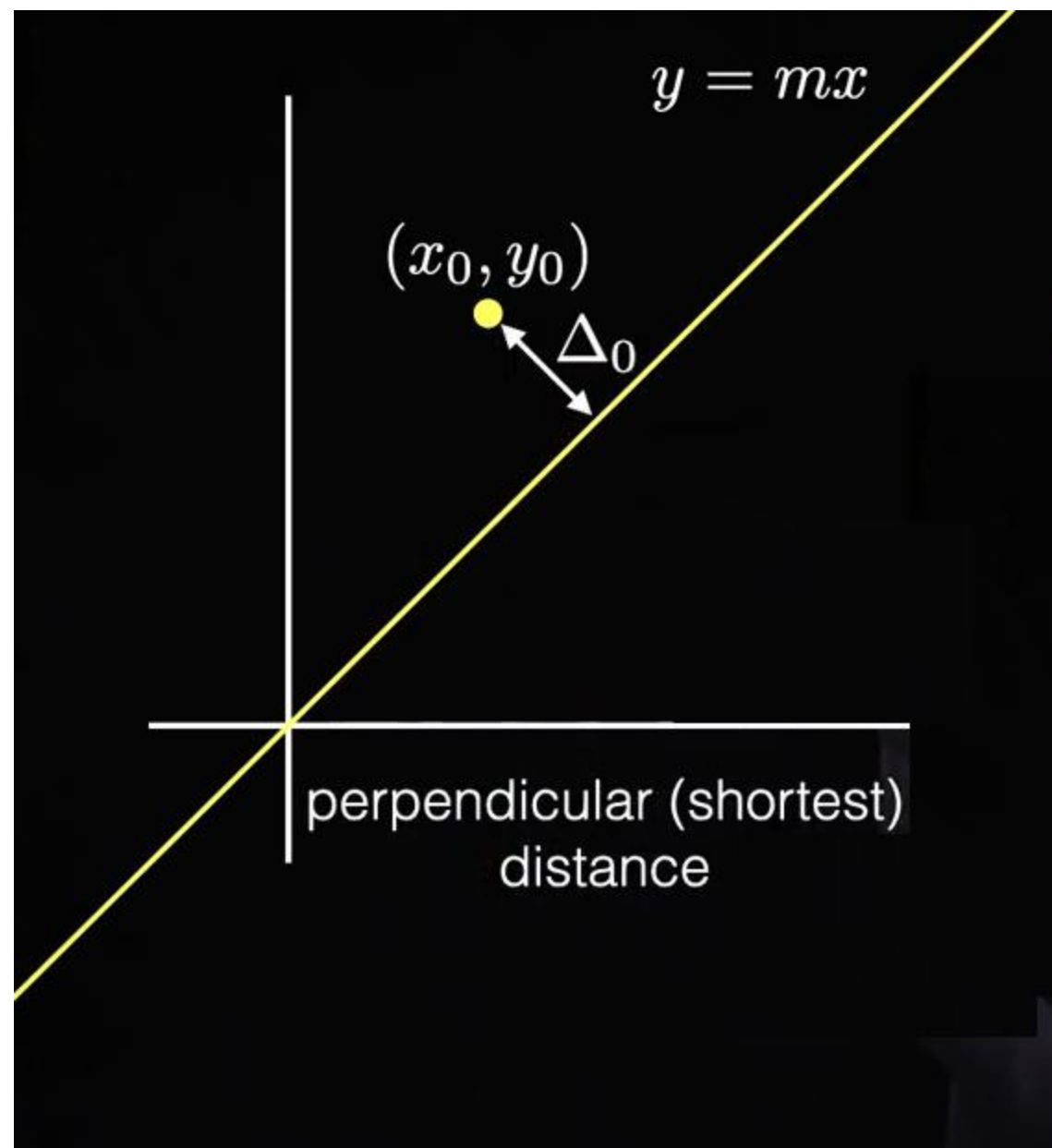
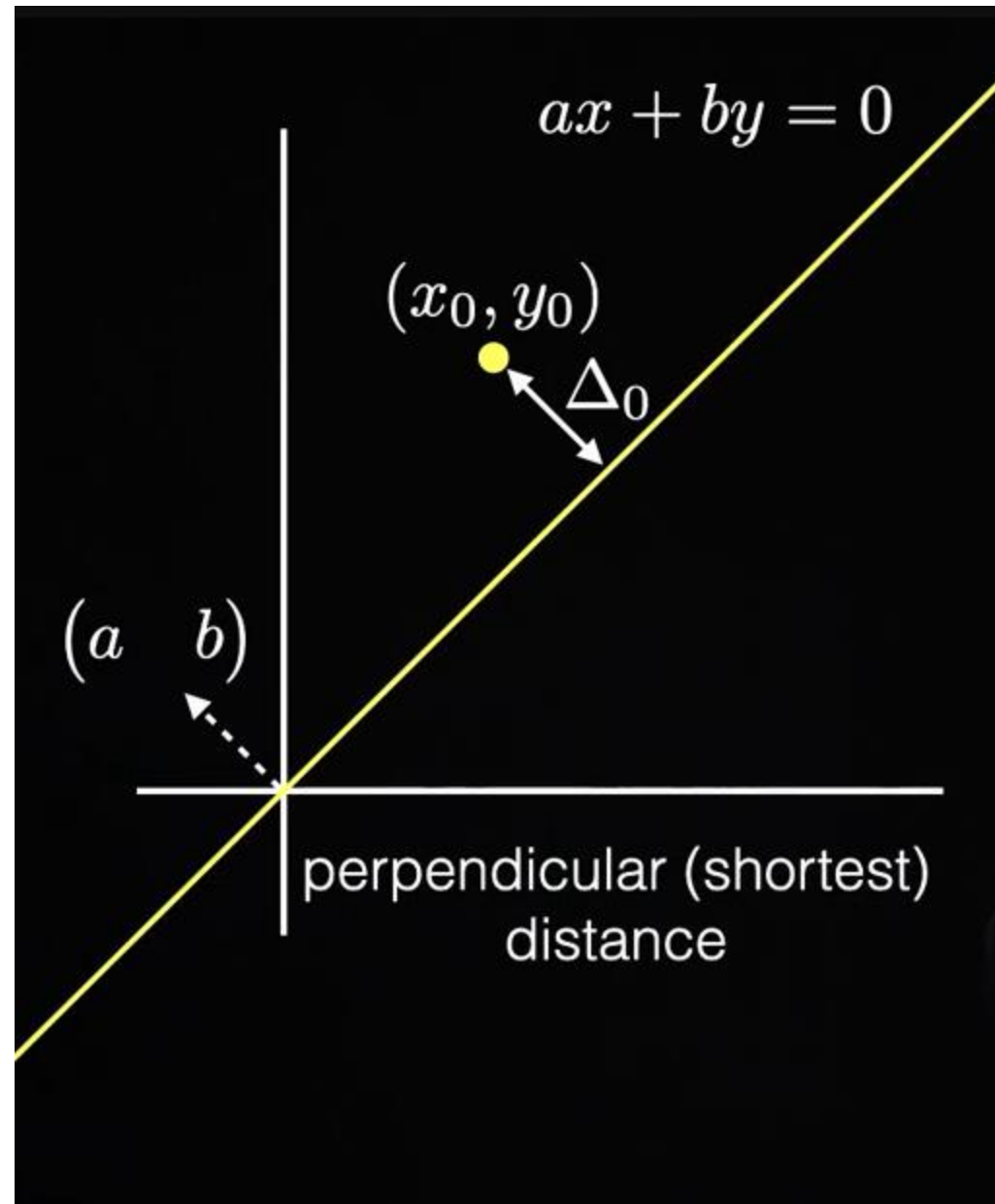
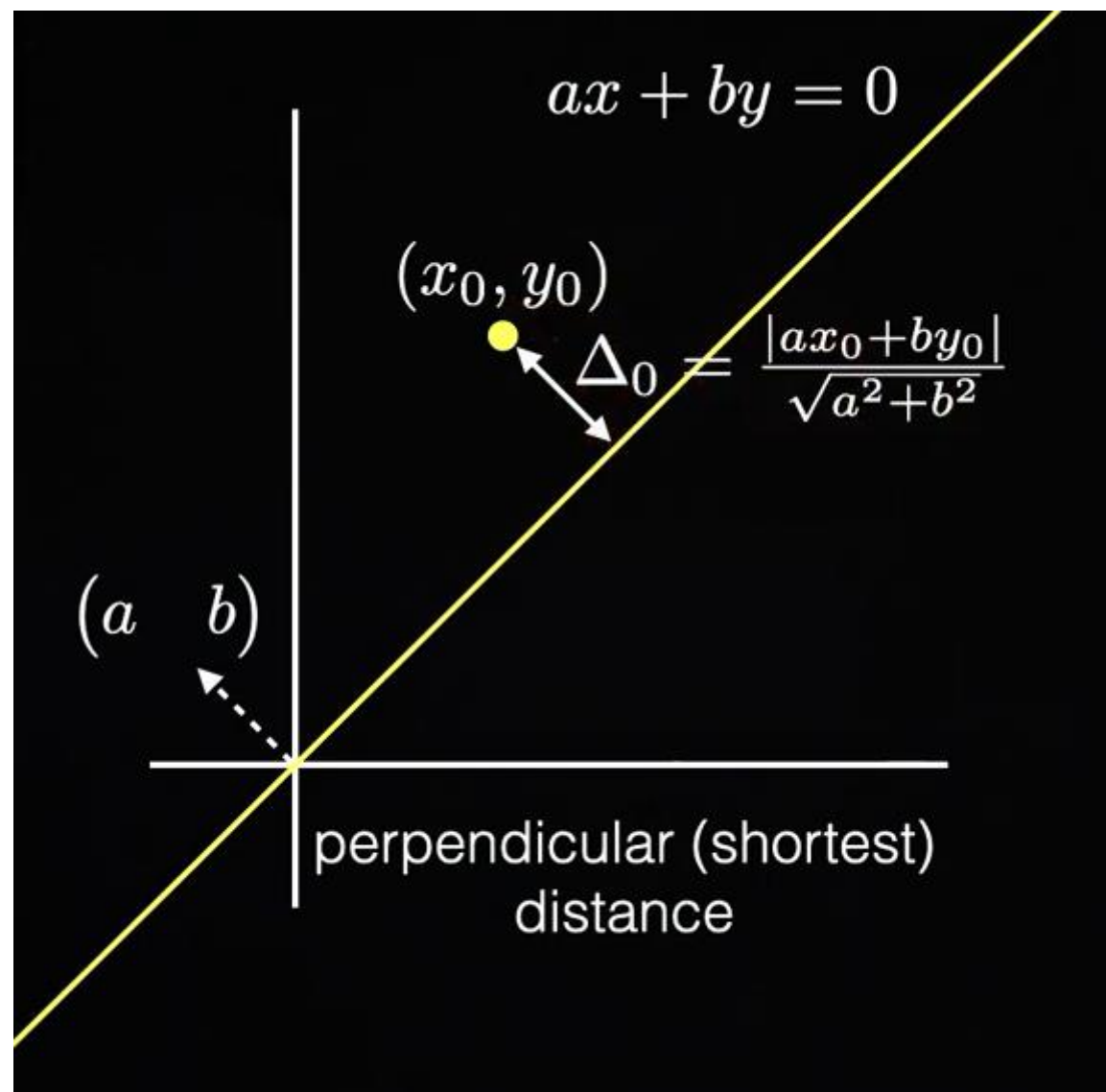


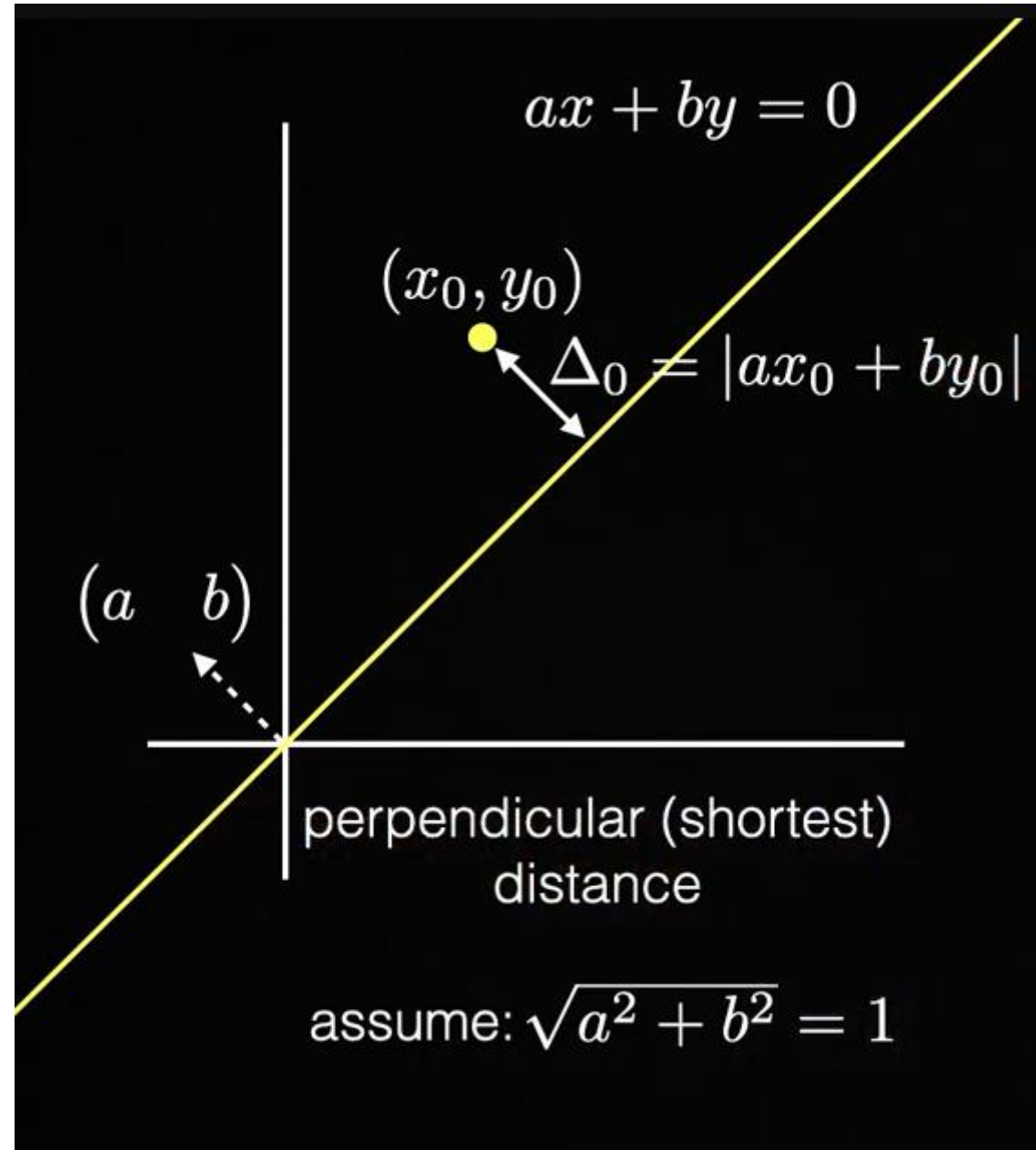
Total least-squares

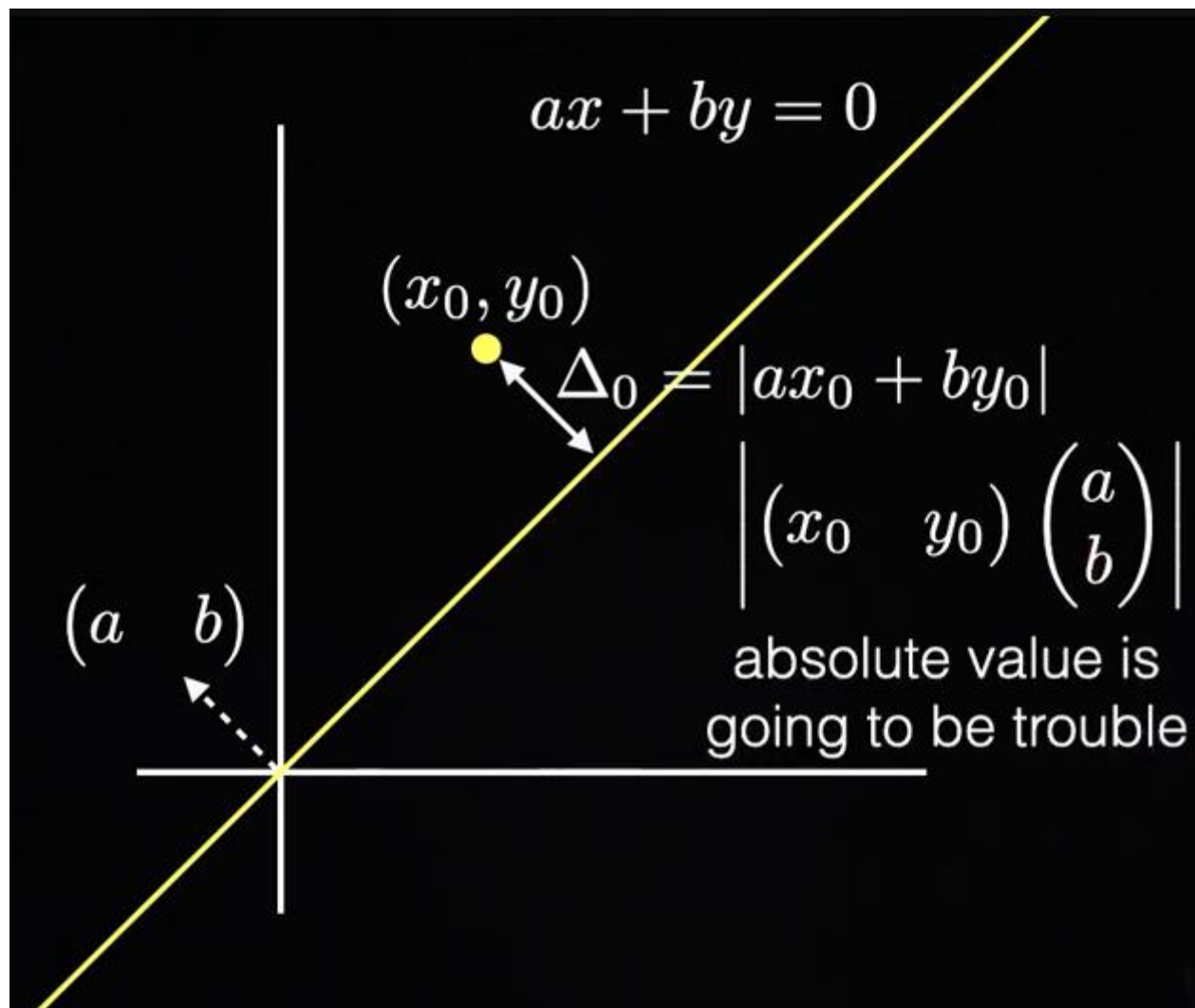


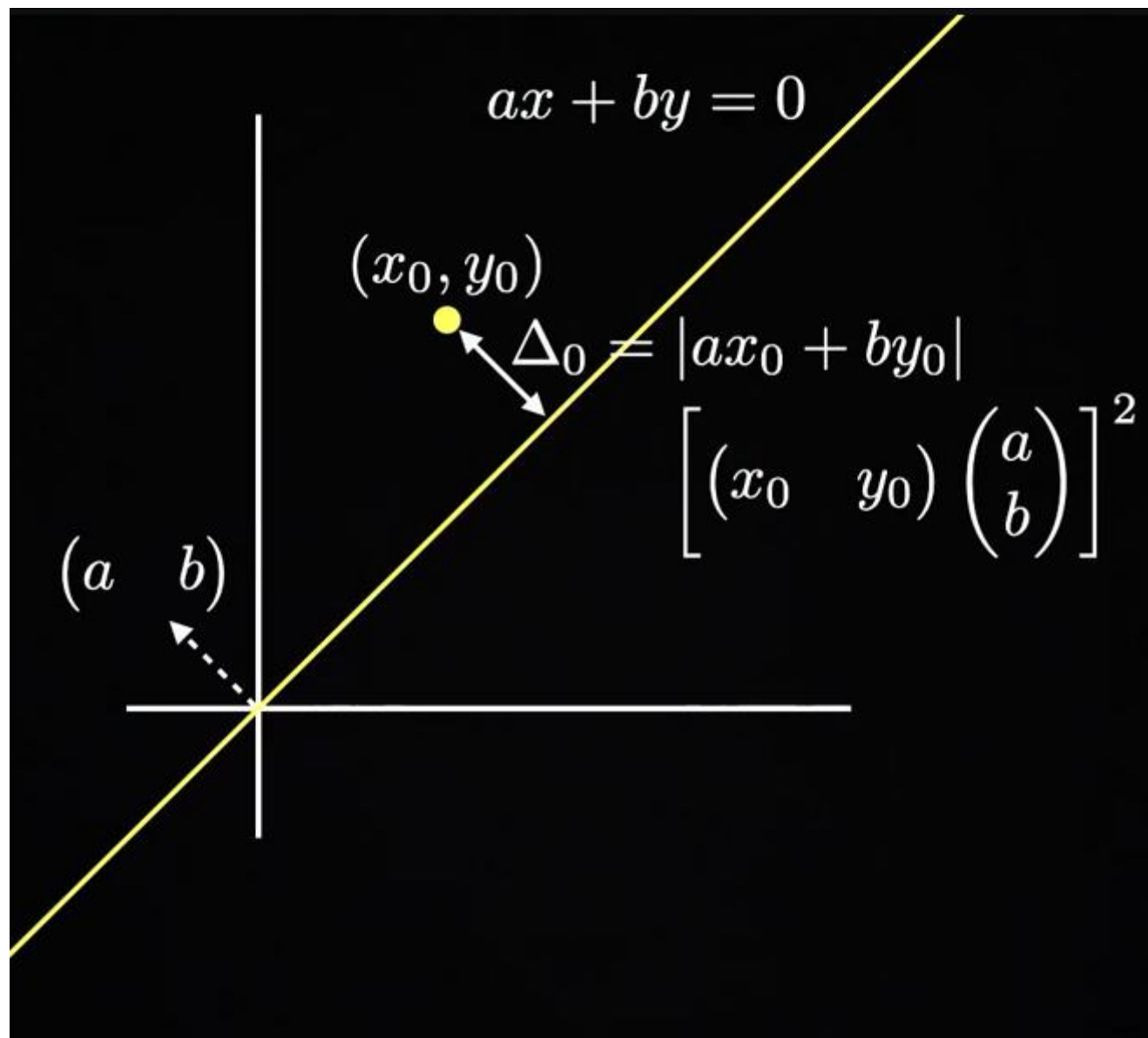


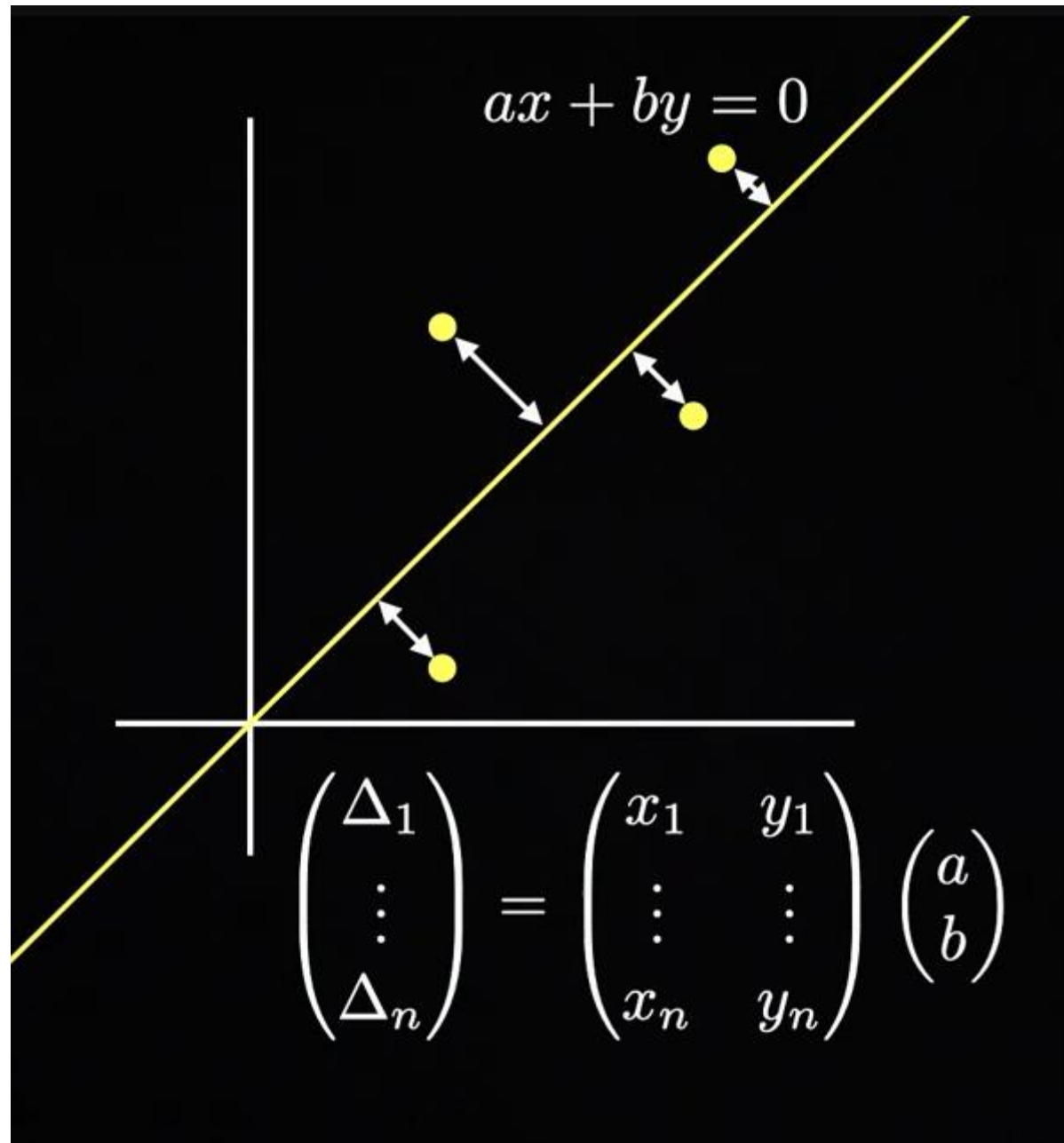


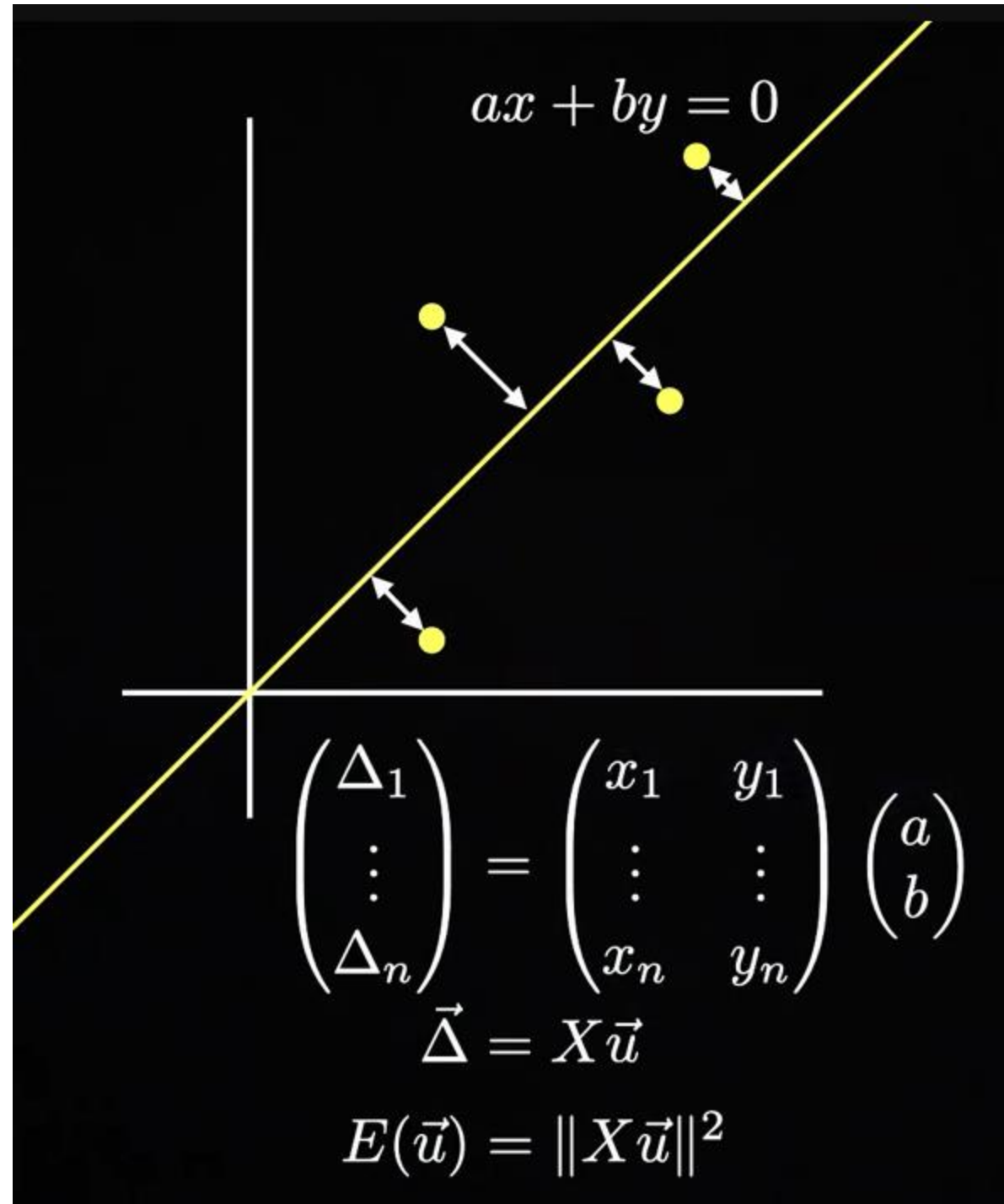








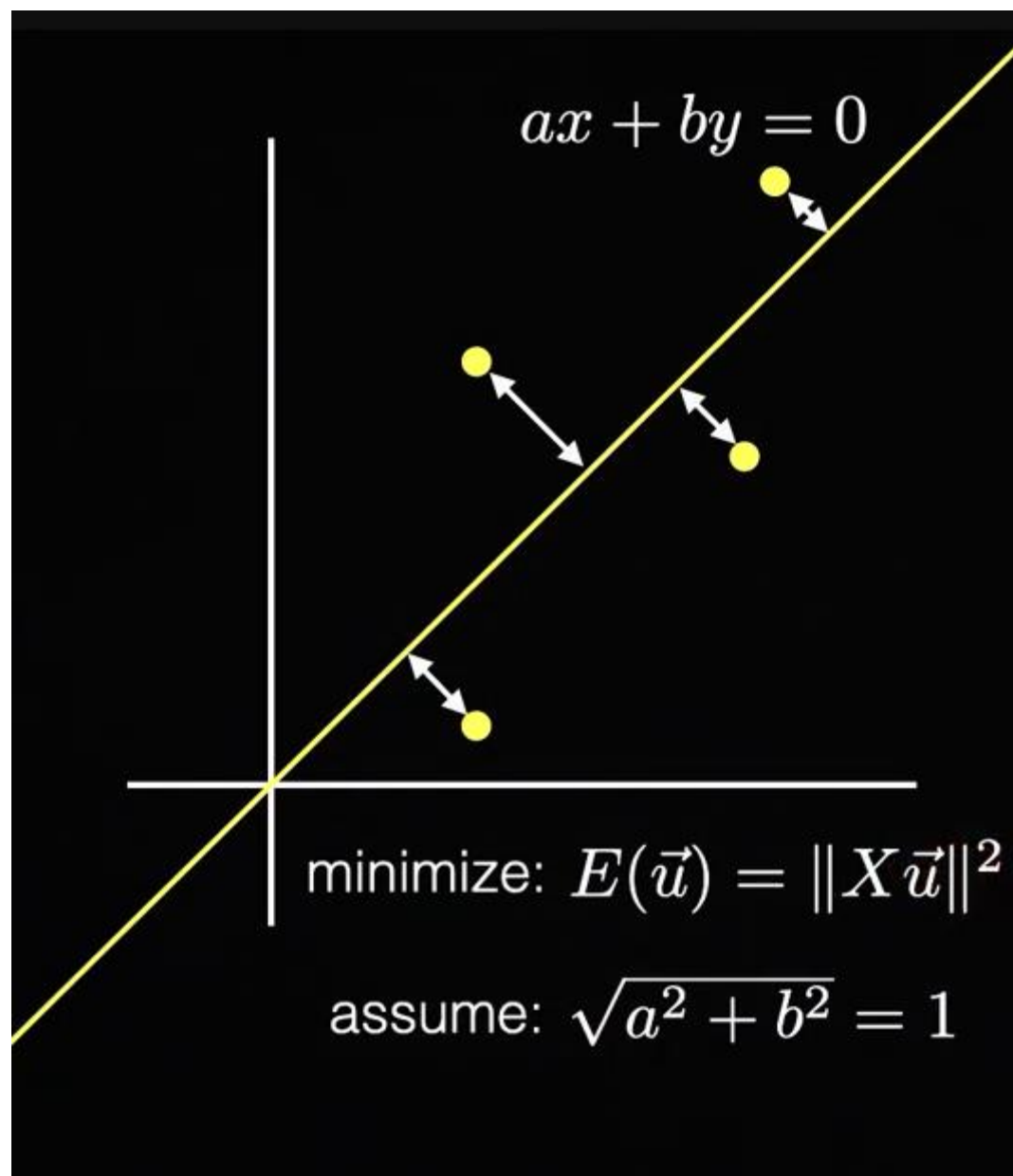


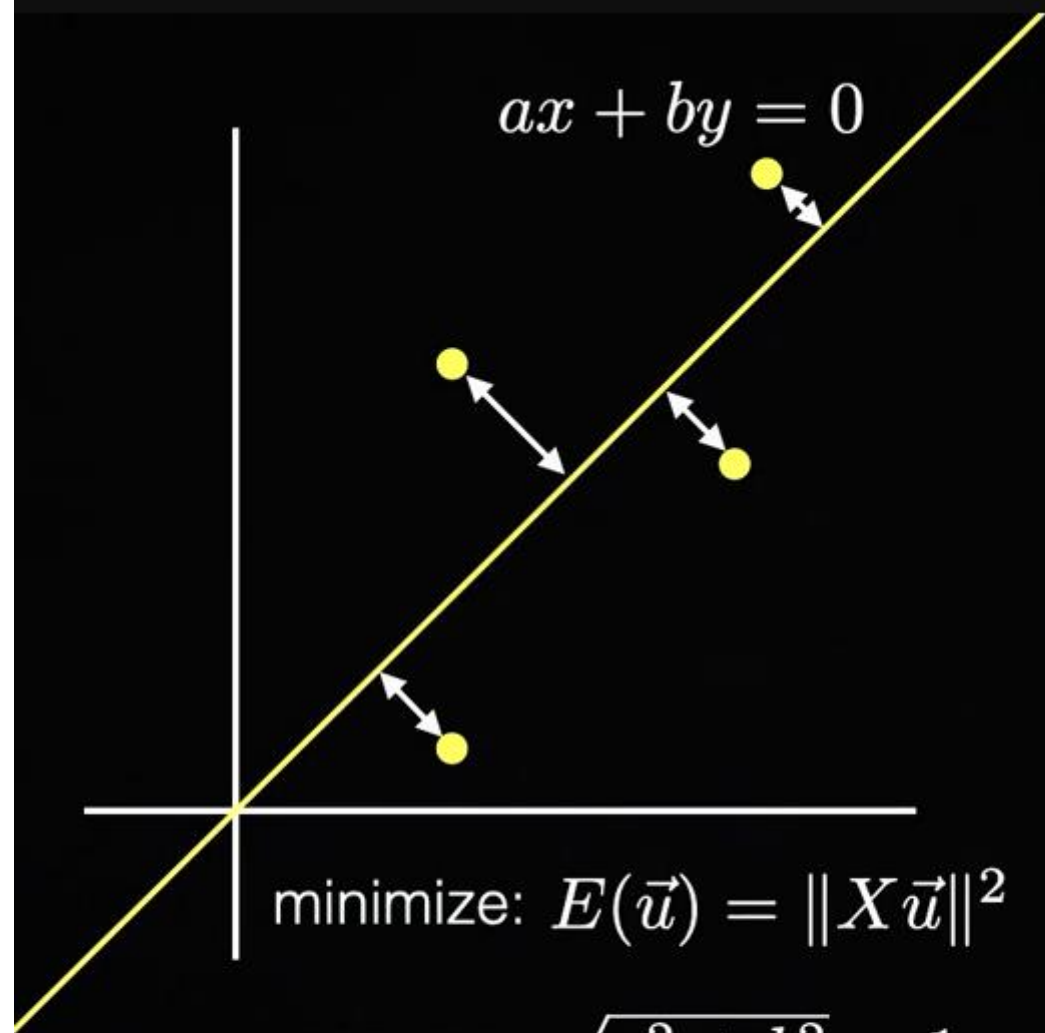


$$\frac{dE}{d\vec{u}} = 2X^T(X\vec{u}) = \vec{0}$$

$$\vec{u} = (X^T X)^{-1} \vec{0}$$

$$\vec{u} = \vec{0} ?$$



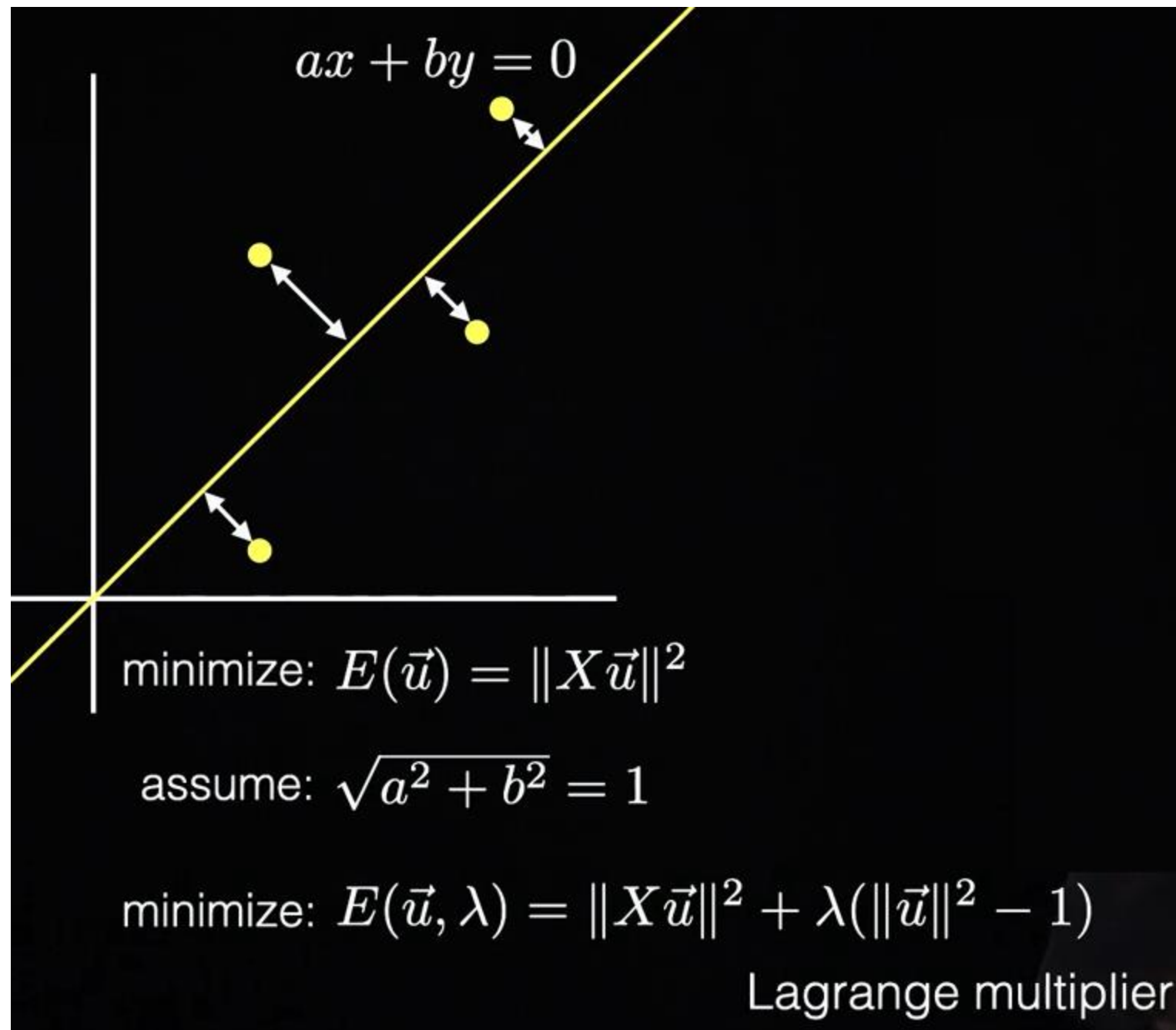


minimize: $E(\vec{u}) = \|X\vec{u}\|^2$

assume: $\sqrt{a^2 + b^2} = 1$

minimize: $E(\vec{u}) = \frac{\|X\vec{u}\|^2}{\|\vec{u}\|^2}$

Rayleigh quotient



$$E(\vec{u}, \lambda) = \|X\vec{u}\|^2 + \lambda(\|\vec{u}\|^2 - 1)$$

$$\frac{dE}{d\vec{u}} = 2X^T(X\vec{u}) + 2\lambda\vec{u} = 0$$

$$(X^T X)\vec{u} = -\lambda\vec{u}$$

$$E(\vec{u}, \lambda) = \|X\vec{u}\|^2 + \lambda(\|\vec{u}\|^2 - 1)$$

$$\frac{dE}{d\vec{u}} = 2X^T(X\vec{u}) + 2\lambda\vec{u} = 0$$

$$(X^T X)\vec{u} = -\lambda\vec{u}$$

$\vec{u}$ is an eigenvector of $(X^T X)$

$$\frac{dE}{d\lambda} = \vec{u}^T \vec{u} - 1 = 0$$

$$\vec{u}^T \vec{u} = 1$$

$$\vec{u}^T (X^T X)\vec{u} = -\lambda \vec{u}^T \vec{u}$$

$$(\vec{u}^T X^T)(X\vec{u}) = -\lambda$$

$$\|X\vec{u}\|^2 = -\lambda$$

$\vec{u}$ is the minimal-eigenvalue eigenvector of $(X^T X)$

Reference

- <https://www.youtube.com/watch?v=zstTZf5AiWE>
- <https://www.youtube.com/watch?v=hgUuPKgTZP8>